RoboEdit: Turning Human Manipulation Videos into Scalable Robot Experience

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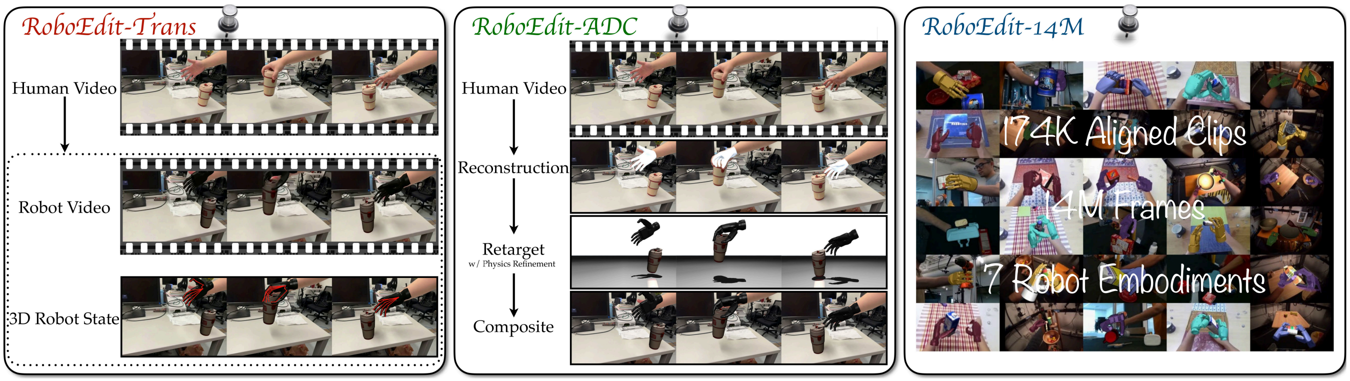

Figure 1: RoboEdit Overview: The framework takes an RGB human video and a target robot, to generate a physically plausible robot interaction video and 3D robot-hand states (RoboEdit-Trans). These states enable motion supervision for downstream robot control, while the RoboEdit-ADC pipeline generates the large-scale paired dataset (RoboEdit-14M) required for training.

Abstract

Collecting robot hand-object interaction data is costly and embodiment-specific, yet abundant human-object videos remain unusable for robot training. We present *RoboEdit*, a human-to-robot video editing suite that transforms human manipulation videos into action-consistent, physically plausible robot videos with aligned 3D hand states. To enable scalable supervision, we introduce *RoboEdit-ADC*, an automatic pipeline that reconstructs and retargets 3D interactions from RGB videos across embodiments. This pipeline generates *RoboEdit-14M*, a large-scale dataset of 174K aligned video pairs (14M frames) spanning seven robot embodiments, diverse scenes, and interaction types. The core editing engine, *RoboEdit-Trans*, employs cross-embodiment adaptation modules to preserve temporal coherence while adapting appearance and motion. It further integrates a 3D Robot-State Decoder to recover per-frame hand states for structured motion supervision. Experiments show that RoboEdit achieves state-of-the-art editing quality and supports downstream robot control policies in real-world manipulation tasks. Ultimately, the RoboEdit suite unlocks the vast potential of unlabeled human videos, providing scalable, high-fidelity visual and 3D motion supervision for generalizable robot learning.

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1 Introduction

Video provides rich supervision for robot manipulation by capturing contact, object motion, viewpoint, scene context, and embodiment appearance—details often lost in compact state vectors (Black et al. 2026). However, collecting robot interaction videos is expensive and strictly embodiment-specific: variations in morphology, kinematics, and control prevent direct data transfer across different robot hands (Zhang et al. 2026). Consequently, existing datasets cover only a fraction of the objects, scenes, viewpoints, and behaviors needed for robust, generalizable robot learning.

Human interaction videos offer a scalable alternative, rich with diverse scenes, viewpoints, object motions, and contact-rich behaviors (Chao et al. 2021; Engel et al. 2023; Liu et al. 2024). Yet embodiment mismatches prevent their direct use as robot training data. While prior work has successfully extracted representations or action supervision from human videos (Nair et al. 2022; Wang et al. 2023), directly translating these videos into physically plausible target-robot interaction videos remains largely unexplored.

This raises a central question: *Can we faithfully transform abundant human videos into scalable, high-fidelity robot interaction videos?* We answer this question through novel human-to-robot manipulation video editing, which retargets the interaction to a target robot while preserving the original

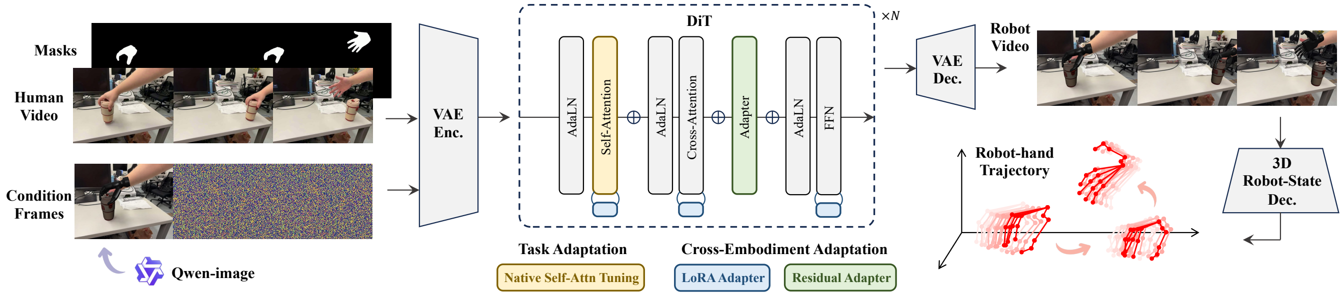

Figure 2: RoboEdit-Trans uses LoRA and residual adapter for editing and 3D Robot-State Decoder for state recovery.

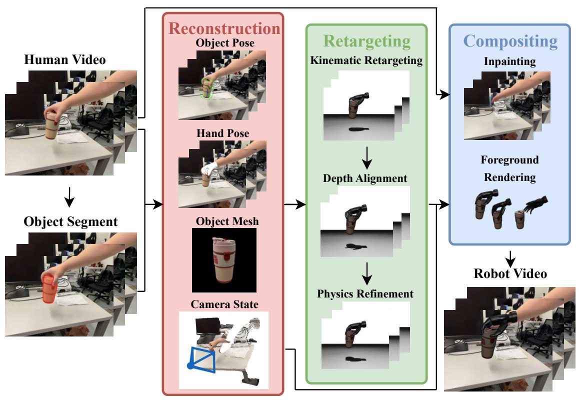

Figure 3: RoboEdit-ADC reconstructs 3D interaction from RGB video, performs depth/physics-refined retargeting, and composites the foreground into the inpainted scene.

scene context and temporal dynamics (Pan et al. 2026b; Jiang et al. 2025; Wei et al. 2026).

We introduce *RoboEdit* (Fig. 1), a suite designed to transform human manipulation videos into scalable, high-fidelity robot training data. It comprises three core components: *RoboEdit-Trans*, a robot-conditioned video editor; *RoboEdit-ADC*, an automatic paired-data curation pipeline; and *RoboEdit-14M*, the resulting large-scale dataset of paired supervision.

Generating entire robot interactions de novo can disrupt the observed scene context and produce physically implausible motion. RoboEdit-Trans (Fig. 2) avoids these pitfalls by editing the RGB human interaction video directly, preserving the source scene, camera motion, and object dynamics while retargeting the human embodiment to the target robot. We introduce cross-embodiment adaptation modules that jointly model cross-embodiment appearance, kinematics, and contact dynamics across diverse robots, ensuring temporal coherence in the generated target-robot video. Furthermore, we integrate a 3D Robot-State Decoder to recover per-frame 3D hand states to serve as structured supervision essential for downstream learning and control.

To provide scalable, aligned supervision, we introduce RoboEdit-ADC (Fig. 3), an automatic pipeline that constructs paired human/robot videos from RGB footage. It first reconstructs the 3D hand-object interaction and camera motion, then retargets the action to a target robot while preserving the original object dynamics and viewpoint. To ensure physical consistency, we incorporate depth regularization and physics-based refinement, effectively reducing artifacts such as object penetration, floating contacts, and temporal jitter. The pipeline concludes by inpainting the human-object regions and compositing the rendered robot interaction into the source scene (Ci et al. 2025; Zi et al. 2025). This process yields RoboEdit-14M (Fig. 4), a massive paired dataset with over 174K aligned video clips (over 14M frames) across 7 robot embodiments, diverse real and synthetic scenes, and varied camera perspectives.

Benchmark comparisons show that RoboEdit-Trans achieves state-of-the-art performance in human-to-robot video editing, outperforming strong baselines in reconstruction fidelity, local editing accuracy, background preservation, and perceptual video quality (He et al. 2025). Ablation studies confirm consistent gains from each component, while real-world deployments validate the utility of the recovered 3D hand states for downstream control. Collectively, these results demonstrate that human-to-robot video editing transforms abundant, unlabeled human manipulation videos into scalable, high-fidelity robot experience, providing both visual and structured trajectory supervision for robot learning.

Our key contributions are as follows:

- We introduce RoboEdit, an end-to-end human-to-robot video editing suite that transforms RGB human manipulation videos into physically plausible robot interaction videos and corresponding structured 3D hand states.
- We develop RoboEdit-Trans, a novel video editor featuring robot-aware adaptation modules for cross-embodiment synthesis and a 3D Robot-State Decoder for 3D robot kinematic recovery.
- We design RoboEdit-ADC, an automatic data curation pipeline that constructs RoboEdit-14M, a large-scale dataset containing 174K aligned clips (14M frames) across 7 robot embodiments, diverse scenes, and interaction types.
- We demonstrate state-of-the-art video editing performance through rigorous benchmarks and ablations, validating the utility of our recovered 3D states for downstream control in real-robot experiments.

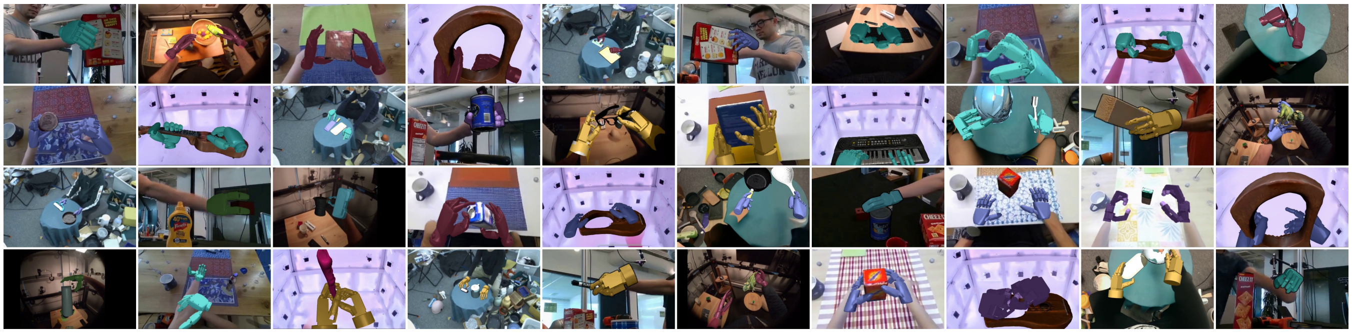

Figure 4: RoboEdit-14M spans diverse everyday manipulation tasks across a wide range of robot embodiments.

2 Related Work

Robot Learning from Human Videos. Human manipulation videos provide a scalable source of interaction data (Grauman et al. 2022; Goyal et al. 2017). Prior work leverages this abundance to pretrain transferable visual representations (Nair et al. 2022; Ma et al. 2023a), learn value functions for imitation and reinforcement learning (Ma et al. 2023b; Zakka et al. 2021), or extract structured interaction priors such as affordances and contact regions (Bahl et al. 2023; Shan et al. 2020). Closer to our work, recent methods treat human hands as action interfaces or reconstruct and retarget hand-object motion into robot trajectories (Shaw, Bahl, and Pathak 2022; Mandikal and Grauman 2022; Sivakumar, Shaw, and Pathak 2022). However, these approaches typically convert human videos into intermediate representations or sparse motion supervision. By contrast, RoboEdit directly generates physically-plausible target-robot videos with corresponding 3D hand states, providing rich visual and kinematic supervision in a single framework.

Human-to-Robot Retargeting. Human-to-robot retargeting adapts human demonstrations to robot embodiments with distinct morphology and kinematics. Classical approaches solve geometric, task-space, or joint-space objectives to align human wrist, fingertip, grasp, or object-relative motion, forming the basis of teleoperation and data collection (Handa et al. 2019; Lakshmipathy, Hodgins, and Pollard 2024). Alternatively, learning-based methods train cross-embodiment policies for direct imitation (Arunachalam et al. 2022; Qin et al. 2022; Shaw, Bahl, and Pathak 2022). For contact-rich manipulation tasks, retargeting further incorporates object dynamics, contact constraints, and physical feasibility through optimization or reinforcement learning (Xin et al. 2025; Pan et al. 2025). Recent monocular pipelines combine 3D hand-object reconstruction with retargeting to derive robot trajectories or policy supervision directly from human videos (Paliwal et al. 2026). However, these methods typically output sparse trajectories, whereas RoboEdit generates full interaction videos alongside dense 3D states.

Human-to-Robot Video Editing. Video editing alters the semantic content of a video while preserving temporal coherence and visual consistency. Diffusion-based approaches have evolved from latent inversion and optimization (Mokady et al. 2023; Qi et al. 2023) to training-free or feed-forward feature manipulation (Geyer et al. 2024; Ku et al. 2024), and increasingly, to multimodal control through references, poses, masks, or trajectories (Liang et al. 2025). Despite the rapid advancement of video diffusion models (Wan et al. 2025), most editing methods assume a fixed agent morphology and motion capabilities. Recent work extends video generation to robotics, synthesizing robot demonstrations from human videos for imitation learning and data generation (Yang et al. 2025; Song et al. 2025). However, human-to-robot video translation presents a unique challenge: it must transform appearance, kinematic structure, and motion dynamics across embodiments while preserving interaction semantics. Recent methods attempt to tackle this using video foundation models and embodiment-aware conditioning (Ci et al. 2025; Xie et al. 2025). Yet significant differences in morphology, kinematics, and physical interaction constraints still limit the accuracy of current cross-embodiment editing.

3 The RoboEdit Suite

Problem Formulation. Given an RGB human hand-object interaction video $v^h_{1:T}$ (where T is the number of frames and h denotes the human domain) and a target robot embodiment e, RoboEdit aims to synthesize a corresponding robot interaction video $\hat{v}^{r,e}_{1:T}$ in the same scene (r denotes the robot domain) and predict the corresponding robot hand state trajectory $\hat{q}^e_{1:T}$.

Since aligned human/robot video pairs are scarce, we utilize RoboEdit-ADC to construct necessary paired supervision. From the source video, the pipeline reconstructs a 3D representation $\mathcal{Z}_{1:T} = (H_{1:T}, O_{1:T}, M_o, m_{1:T}, C_{1:T})$, where H and O are the human hand and object pose trajectories, M_o is the object mesh, m is the edit mask, and C represents the source camera states. The pipeline then retargets the human trajectory $H_{1:T}$ relative to the object $O_{1:T}$ into a robot-hand state trajectory $q^e_{1:T}$, inpaints the human hand and object regions, and renders the robot-object interaction under the original camera motion $C_{1:T}$ to produce the target video $v^{r,e}_{1:T}$.

Each curated sample is defined as

$$\mathcal{D}_i = (v^h_{1:T},\, m_{1:T},\, v^{r,e}_{1:T},\, q^e_{1:T}).$$

These samples constitute the RoboEdit-14M dataset (detailed in Section 3.3). Finally, RoboEdit-Trans learns a mapping $F_\theta(v^h_{1:T}, e) \rightarrow (\hat{v}^{r,e}_{1:T}, \hat{q}^e_{1:T})$, preserving the source scene, camera motion, and object dynamics while transforming the human interaction to the target embodiment e.

3.1 RoboEdit-ADC: Automatic Data Curation

3D Interaction Reconstruction. As illustrated in Fig. 3, RoboEdit-ADC reconstructs the full 3D human-object interaction from a single RGB video. We estimate the articulated hand trajectory $H_{1:T}$ using HaMeR (Pavlakos et al. 2024) and segment the manipulated object using SAM 2 (Ravi et al. 2024). The edit masks $m_{1:T}$ are then derived by combining the object segmentation with the hand mask. For object geometry, we employ TRELLIS (Xiang et al. 2024) to reconstruct the mesh M_o and FoundationPose (Wen et al. 2024) to track its 6D pose trajectory $O_{1:T}$. Finally, VGGT (Wang et al. 2025) estimates the source camera state $C_{1:T}$, including intrinsics, extrinsics, and depth. To ensure physical consistency, we rectify the reconstruction via coordinate-frame and scale calibration before retargeting (see Supplement Sec. A.1).

Monocular HaMeR estimates often suffer from depth and scale ambiguities, such that even an image-aligned hand can be misplaced in 3D space, leading to missed contacts, object penetration, or instability after retargeting. To address this, we leverage the aligned depth to correct HaMeR's camera-space scale and depth prior to retargeting. We project the recovered palm joints into the depth map and back-project the depth observations into metric 3D palm anchors to estimate a scale factor. We then rescale the camera-space hand reconstruction about the camera center to align its palm depth with the depth observations, while keeping the relative hand articulation fixed. This correction improves the hand-object contact geometry for subsequent retargeting. Finally, the reconstruction stage outputs the unified hand-object-camera representation:

$$\mathcal{Z}_{1:T} = (H_{1:T}, O_{1:T}, M_o, m_{1:T}, C_{1:T})\,.$$

Human-Robot Retargeting. Given $\mathcal{Z}_{1:T}$ and a target embodiment e, RoboEdit-ADC retargets the human hand motion using a strategy inspired by SPIDER (Pan et al. 2025). For anthropomorphic hands, we use SPIDER's wrist-and-fingertip IK in the MuJoCo (Todorov, Erez, and Tassa 2012) simulator. For morphologically distinct two- and three-finger grippers, we apply specialized strategies: a two-finger gripper derives its jaw pose and opening from wrist and thumb-index geometry, while a three-finger gripper estimates palm pose and optimizes joint angles to match the thumb, index, and middle fingertips. This yields an initial kinematic trajectory $q^{e,\mathrm{IK}}_{1:T}$.

To mitigate penetration, floating contacts, and motion artifacts caused by reconstruction noise and morphological mismatches, we refine this trajectory via a physics-guided refinement: $q^e_{1:T} = \arg\min_{\{q^e_t \in \mathcal{Q}_e\}_{t=1}^{T}} \mathcal{L}_{\text{ret}}$, where $\mathcal{Q}_e$ denotes the set of joint-limit-feasible configurations for e and

$$\mathcal{L}_{\text{ret}} = \lambda_{\text{track}}\mathcal{L}_{\text{track}} + \lambda_{\text{geo}}\mathcal{L}_{\text{geo}} + \lambda_{\text{contact}}\mathcal{L}_{\text{contact}} + \lambda_{\text{temp}}\mathcal{L}_{\text{temp}}.$$

The terms of this objective function are defined as follows:
Tracking Loss: $\mathcal{L}_{\text{track}} = \sum_{t=1}^{T} \|q^e_t - q^{e,\mathrm{IK}}_t\|_2^2$ ensures fidelity to the kinematic reference.
Temporal Loss: $\mathcal{L}_{\text{temp}} = \sum_{t=2}^{T} \|q^e_t - q^e_{t-1}\|_2^2$ smooths motion and reduces jitter.
Geometry Loss: $\mathcal{L}_{\text{geo}}$ is an L_2 penalty on robot-object penetration detected via MuJoCo collision checks and visual-mesh geometry (see Supplement Sec. A.2).
Contact Loss: $\mathcal{L}_{\text{contact}} = \sum_{t=1}^{T} \sum_{k \in \mathcal{C}_t} \|p^e_k(q^e_t) - a_{t,k}\|_2^2$ preserves valid contacts, where $\mathcal{C}_t$ is the set of fingertip indices whose reconstructed fingertip-to-surface distance is below a threshold consistently across neighboring frames, $a_{t,k}$ is the closest 3D object-surface point for each $k \in \mathcal{C}_t$, and $p^e_k(q^e_t)$ is the 3D position of robot fingertip obtained from FK $p^e_k(\cdot)$.

The optimized trajectory $q^e_{1:T}$ is then used for rendering the target robot interaction.

Robot Interaction Compositing. Given the source video $v^h_{1:T}$, edit masks $m_{1:T}$, robot and object trajectories $q^e_{1:T}$ and $O_{1:T}$, and aligned camera states $C_{1:T}$, RoboEdit-ADC constructs the paired robot video in three steps: (1) MiniMax-Remover (Zi et al. 2025) inpaints the human hand and object regions within $m_{1:T}$, producing a clean background $b_{1:T}$; (2) we render the robot hand and object under C_t as foreground r_t; and (3) we composite r_t onto b_t to obtain the target frame $v^{r,e}_t$ and the full video $v^{r,e}_{1:T}$. This process preserves the source background and camera motion while seamlessly substituting the hand-object interaction with the target robot.

3.2 RoboEdit-Trans: Human-Robot Video Editing

Model Overview and Robot Conditioning. RoboEdit-Trans extends the NovaEdit (Pan et al. 2026b) architecture to enable cross-embodiment robot editing and 3D robot-state prediction (Fig. 2). We introduce adaptation modules supporting diverse robot embodiments and a 3D Robot-State Decoder to recover the robot hand trajectory. The editor takes a masked human video, providing the scene, camera motion, and object dynamics, and sparse target-robot condition frames specifying the target embodiment and representative hand-object configurations. Let z^h and $z^{c,e}$ denote their respective latent representations. The editor predicts target-robot latent $\hat{z}^{r,e} = G_\theta(z^h, z^{c,e})$, where G_θ is the robot-conditioned editor (c denotes conditioning) optimized via flow matching.

Cross-Embodiment Adaptation. The shared editor learns a common human-to-robot task adaptation, but variations in robot appearance, morphology, kinematics, and contact patterns require cross-embodiment adaptation. We introduce two complementary modules: LoRA adapters (Hu et al. 2021) adapt spatiotemporal representations to diverse robot appearance and motion, while residual bottleneck adapters refine embodiment-specific hand geometry and interaction patterns. For input feature x and intermediate feature h, LoRA and residual-adapter mappings f_{LoRA} and f_{Ada} are:

$$f_{\text{LoRA}}(x) = W_0 x + \tfrac{\alpha}{r} BAx,$$
$$f_{\text{Ada}}(h) = h + W_2^{\text{Ada}} \phi\big(W_1^{\text{Ada}} \text{LayerNorm}(h)\big)\,.$$

Here, W_0 is the frozen pretrained weight, A and B are trainable rank-r matrices, and α scales the LoRA update. W_1^{Ada} and W_2^{Ada} are the trainable down- and up-projection matrices; LayerNorm denotes layer normalization, and ϕ is a nonlinear activation. Together, these modules adapt one shared editor across target embodiments while retaining its human-to-robot editing capability.

3D Robot-State Decoder. Edited videos do not explicitly encode metric 3D motion, so we leverage a 3D Robot-State Decoder to recover camera-space robot states. Our 3D Robot-State Decoder first estimates robot states in each frame and then refines them over the full sequence. For framewise

Dataset	Frames	View	Camera	Scene	Emb.	RGB Pair	Robot State	Auto
H&R (Xie et al. 2025)	$\sim$1.2M	Third	Static	Real	1	✓	✓	×
H2R (Li et al. 2026)	$\sim$1M	Ego	Moving	Real	3	✓	×	✓
UniDex (Zhang et al. 2026)	9M	Ego	Moving	Real	8	×	✓	×
X-Humanoid (Yang et al. 2025)	2.8M	Third	Moving	Synth.	1	✓	×	×
RoboEdit-14M (Ours)	14.1M	Ego+Third	Both	Both	7	✓	✓	✓

Table 1: Comparison of representative human-to-robot manipulation datasets. Frames follow each work's reported scale; Emb. denotes the number of robot embodiments; Auto denotes curation without per-sample human intervention.

Figure 5: Qualitative comparison across four human-object interactions and four robot embodiments. The left two columns show RoboEdit-ADC results, while the remaining columns compare RoboEdit-Trans with baseline video editing models.

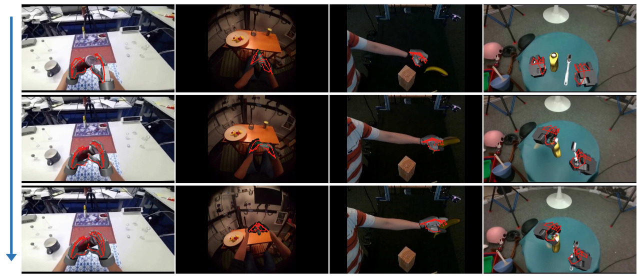

Figure 6: 3D robot-state predictions on RoboEdit-Trans edited videos across four embodiments.

spatial estimation, inspired by heatmap-based 2D hand pose estimation (Iqbal et al. 2018), we use a shared mask-aware image encoder (He et al. 2016) and design robot-specific prediction heads to estimate per-frame 2D palm anchors and fingertips, palm-frame fingertip coordinates, and joint configurations across diverse embodiments. A shared camera head predicts camera intrinsics, while the predicted 2D palm anchors are matched with embodiment-specific 3D palm geometry to recover the 3D wrist pose through Perspective-n-Point (PnP). A temporal Transformer then jointly refines the framewise states over the full sequence to enforce temporal consistency and reduce frame-to-frame jitter, after which forward kinematics produces the complete camera-space robot-hand trajectory. The decoded states provide structured motion guidance for downstream robot learning and control.

We supervise per-frame 2D localization, 3D geometry, articulation, and camera estimation together with temporal state consistency. More details are in Supplement Sec. B.

3.3 RoboEdit-14M: Paired Human-Robot Dataset

Dataset Overview. As shown in Fig. 4, RoboEdit-14M contains 174,547 aligned human/robot video pairs totaling over 14.1M paired frames, constructed from 24,197 human-interaction clips. Each pair includes a target robot video and its retargeted 3D hand trajectory, providing large-scale visual and robot-state supervision. Table 1 shows that RoboEdit-14M uniquely combines automatic curation, paired RGB video, and robot-state labels at 14.1M-frame scale. More details are in Supplement Sec. C.

Source Diversity. We source human videos from DexYCB (Chao et al. 2021), HOT3D (Banerjee et al. 2024), H2O (Kwon et al. 2021), GigaHands (Fu et al. 2024), and TACO (Liu et al. 2024), covering static and moving egocentric views, single- and bimanual manipulation, and diverse objects and tools. RoboEdit-ADC converts them into 145,459 real-scene pairs spanning 7 embodiments: Inspire, XHand, Ability, SCHUNK SVH, Allegro, Unitree Dex3, and the Franka Panda gripper.

Synthetic Scene Augmentation. To expand visual diversity, we construct 29,088 synthetic pairs by rendering aligned human and retargeted scene under varied cameras and lighting. We composite both foregrounds onto the same background generated by RoboEngine (Yuan et al. 2025), preserving source–target alignment while diversifying scene context.

4 Experiments

4.1 Experimental Setup

Training and Inference Details. RoboEdit-Trans builds on NovaEdit's Wan2.1-VACE-1.3B backbone (Wan et al. 2025; Jiang et al. 2025). We fine-tuned the backbone on paired human/robot clips for task adaptation, then froze it and trained LoRA and residual adapters jointly for cross-embodiment adaptation. Following NovaEdit, we used 81-frame clips with sparse target-robot keyframes at indices $\{0, 10, \ldots, 80\}$. At inference, we used Qwen-Image-Edit (Wu et al. 2025), fine-tuned on RoboEdit-14M, to generate these keyframes; RoboEdit-Trans then used them to produce the edited video.

We trained the 3D Robot-State Decoder in two stages: the framewise spatial estimator on rendered and composited robot frames, followed by the temporal Transformer on full 81-frame sequences with the spatial estimator frozen. At inference, it recovered the camera-space robot-hand trajectory from the generated video. Supplement Sec. D provides the details.

Benchmark and Metrics. We evaluated RoboEdit-Trans on a fixed benchmark of 300 cases spanning diverse source domains, cameras, interactions, and embodiments. We report SSIM (Wang et al. 2004) and LPIPS (Zhang et al. 2018) against the paired reference, edit-region LPIPS against the same reference, and background SSIM against the source outside the edit mask. Additionally, we report VBench (Huang et al. 2024) measuring visual quality and temporal consistency, and OpenVE-Bench local-replacement protocol (He et al. 2025) assessing prompt compliance, visual naturalness, temporal stability, and physical and motion integrity.

Baselines and Protocols. We compared RoboEdit-Trans with VACE (Jiang et al. 2025), UniVideo (Wei et al. 2026), VINO (Chen et al. 2026), Kiwi-Edit (Lin et al. 2026), OmniWeaving (Pan et al. 2026a), EditCtrl (Litman et al. 2026), AnyV2V (Ku et al. 2024), and ReCo (Zhang et al. 2025). Following each method's supported interface, single-reference methods received one robot frame, while multi-reference methods were also evaluated with sparse robot keyframes.

4.2 Qualitative Results

Fig. 5 and Fig. 6 show 4 cases from distinct sources and robot hands, using only midpoint non-keyframes such as 15 and 25 to best show editing and 3D decoding ability. Fig. 5 compares all methods at matched timestamps and crops. RoboEdit-Trans consistently synthesized the target embodiment while preserving the scene and object motion, whereas the baselines often retained human hands or generated inconsistent hand geometry. Fig. 6 shows that the 3D Robot-State Decoder tracks the generated robot hands and recovered their corresponding 3D motion. More results are in Supplement Sec. F.

4.3 Quantitative Comparison

Table 2 shows that RoboEdit-Trans achieves SOTA performance, with stronger reconstruction and local-editing while remaining competitive in visual quality and motion consistency. Its slightly lower BG SSIM reflects the differing spatial extent of robot and human hands, which causes valid changes near the edit-mask boundary to count as background errors.

Method	Params	Condition	Reconstruction		Local Editing		VBench			OpenVE
			SSIM ↑	LPIPS ↓	Edit LPIPS ↓	BG SSIM ↑	AQ ↑	DD ↑	MS ↑	Overall ↑
VACE	1.3B	Single reference	0.8764	0.1070	0.0497	**0.9487**	0.4673	0.4867	0.9952	3.2270
UniVideo	14B	Single reference	0.3249	0.6515	0.1043	0.4026	0.3871	0.4133	0.9890	2.2455
VINO	13B	Single reference	0.5564	0.3348	0.0679	0.6041	0.4488	0.4433	**0.9967**	**3.2711**
Kiwi-Edit	5B	Single reference	0.7415	0.2059	0.0627	0.8137	0.4307	0.4600	0.9950	3.2386
OmniWeaving	13B	Single reference	0.8107	0.1590	0.0625	0.8855	0.4478	**0.6900**	0.9955	3.1634
EditCtrl	1.3B	Text instruction	0.5452	0.4295	0.0809	0.6154	0.4319	0.4933	0.9951	3.0319
AnyV2V	1.3B	Single reference	0.5060	0.4837	0.0978	0.5753	0.3891	0.1567	0.9756	2.9023
ReCo	1.3B	Single reference	0.8433	0.1366	0.0546	0.8869	0.4827	0.4967	0.9939	3.1255
VACE	1.3B	Multi-keyframe	0.8996	0.0778	0.0258	0.9419	0.4768	0.5667	0.9923	3.1446
UniVideo	14B	Multi-keyframe	0.3364	0.6464	0.1032	0.4125	0.3836	0.4233	0.9900	2.2089
VINO	13B	Multi-keyframe	0.5539	0.3380	0.0541	0.5897	0.4396	0.2733	0.9944	3.1447
Ours	1.3B	Multi-keyframe	**0.9282**	**0.0470**	**0.0171**	0.9188	**0.4832**	0.6300	0.9956	3.2511

Table 2: Quantitative results on the 300-case RoboEdit-Trans benchmark. AQ/DD/MS denote VBench Aesthetic Quality/Dynamic Degree/Motion Smoothness; OpenVE is the overall OpenVE-Bench score. (Best results: bold. Second-best: underlined.)

LoRA	Adapter	SSIM ↑	LPIPS ↓	Edit LPIPS ↓	BG SSIM ↑
×	×	0.9255	0.0494	0.0179	0.9174
✓	×	0.9264	0.0490	0.0176	0.9181
×	✓	0.9276	0.0478	0.0173	0.9186
✓	✓	**0.9282**	**0.0470**	**0.0171**	**0.9188**

Table 3: Ablation of adaptation modules on 300 cases using identical sparse-keyframe conditioning.

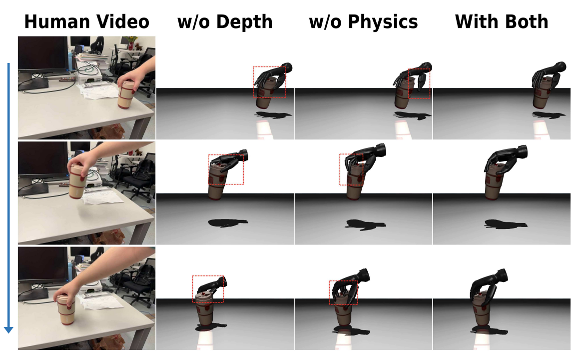

Figure 7: Qualitative ablation of RoboEdit-ADC retargeting. Depth regularization improves hand-object alignment, while physics refinement reduces floating and penetration.

4.4 Ablation Study

Video Editing Components. Table 3 reports the contributions of the two cross-embodiment adaptation modules. Both LoRA and the residual adapter improve all metrics over the backbone-only variant; the residual adapter provides the larger individual gain, while combining both performs best.

Retargeting Quality. Fig. 7 presents the two refinements in RoboEdit-ADC. Without depth regularization, monocular depth and scale errors propagate to retargeting and displace the robot hand in the scene. Without physics refinement, kinematic matching produces floating or penetrating fingers.

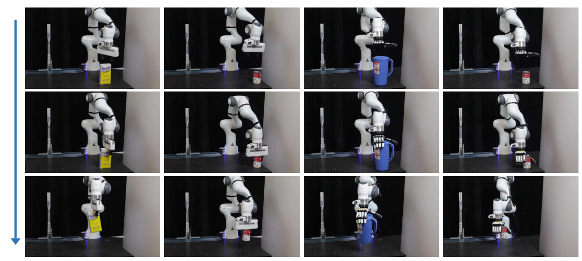

Figure 8: Real-robot deployment with 3D Robot-State Decoder trajectories across four YCB-object tasks.

4.5 Real-Robot Deployment

To evaluate our 3D Robot-State Decoder for downstream control, we trained a residual PPO controller (Schulman et al. 2017) in Genesis (Genesis Authors 2024) to track robot-hand trajectories decoded from edited videos. Across 512 randomized simulation environments, it achieves trajectory-reproduction success rates of 71% with the Panda gripper and 62% with XHand. We further deployed the controller on a 7-DoF Franka Panda for YCB-object manipulation (Fig. 8), demonstrating that decoded 3D trajectories support successful real-robot execution. More details are in Supplement Sec. E.

5 Conclusions

We introduced RoboEdit, a comprehensive suite that transforms human manipulation videos into physically plausible robot interaction videos and structured 3D hand states. RoboEdit-ADC automatically curates the RoboEdit-14M dataset to provide scalable paired supervision, while RoboEdit-Trans performs cross-embodiment editing and 3D robot state decoding. Our experiments demonstrate state-of-the-art editing quality, consistent gains from each component, and effective motion guidance for downstream control. Collectively, they prove that RoboEdit unlocks the vast potential of abundant human videos, converting them into scalable, high-fidelity robot experience for generalizable learning.

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Supplementary Material

A RoboEdit-ADC Details

A.1 Camera Alignment and Depth Regularization

Camera Alignment. VGGT predicts camera-from-world extrinsics $E_t = [R_t^{\mathrm{cw}} \mid t_t^{\mathrm{cw}}]$, camera intrinsics K_t, and depth maps D_t. We convert each extrinsic into camera center $c_t = -R_t^{\mathrm{cw}\top} t_t^{\mathrm{cw}}$ and orientation $R_t^{\mathrm{wc}} = R_t^{\mathrm{cw}\top}$. Because the VGGT reconstruction has an arbitrary scale and world frame, we align it to the hand-object rendering frame with a global similarity transform (s, R_a, t_a). We estimate this transform using standard closed-form SVD alignment of sparse corresponding camera centers between the VGGT and rendering coordinate systems. The aligned camera and depth are $c'_t = sR_a c_t + t_a$, $R'_t = R_a R_t^{\mathrm{wc}}$, and $\widetilde{D}_t = sD_t$, yielding $C_t = (K_t, c'_t, R'_t)$.

Depth Regularization. For depth regularization, we use the aligned depth map to correct the scale and depth of the monocular HaMeR hand reconstruction before retargeting. We select stable wrist and finger-base landmarks from MANO (Romero, Tzionas, and Black 2017) as palm anchors and denote their set by $\mathcal{A}$. For anchor j, we project its HaMeR camera-space position $H_{t,j}$ with K_t, map the resulting pixel $\mathbf{u}_{t,j} = (u_{t,j}, v_{t,j})$ to the resolution of $\widetilde{D}_t$, and take the median positive, finite depth $d_{t,j}$ within a 5×5 neighborhood. We then back-project this observation into a metric 3D anchor:

$$p_{t,j}^D = d_{t,j} K_t^{-1} \left[u_{t,j}, v_{t,j}, 1\right]^\top.$$

Invalid samples are excluded. From the valid set $\mathcal{A}_t$, we compute $\bar{z}_t^D = \mathrm{median}_{j \in \mathcal{A}_t} (p_{t,j}^D)_z$ and $\bar{z}_t^H = \mathrm{median}_{j \in \mathcal{A}_t} (H_{t,j})_z$, giving $\alpha_t = \bar{z}_t^D / \bar{z}_t^H$. We rescale the complete camera-space hand about the camera center as $\widetilde{H}_t = \alpha_t H_t$, applying the same factor to its global translation while leaving the MANO pose, shape, and relative articulation unchanged. This camera-centered isotropic scaling aligns the metric palm depth while preserving the original image projection.

A.2 Physics-guided Refinement

Refinement Objective. For each retargeted trajectory, we initialize from the kinematic trajectory $q_{1:T}^{e,\mathrm{IK}}$ and optimize the complete temporal window under its kinematics, joint limits, palm geometry, and fingertip correspondences. We retain only trajectories that respect robot joint limits and maintain consistent hand-object contact over time. The main paper defines $\mathcal{L}_{\mathrm{ret}}$; here we expand its four terms. Let $[x]_+ = \max(x, 0)$.

$$\begin{aligned}
\mathcal{L}_{\mathrm{track}} &= \sum_{t=1}^{T} \left\| q_t^e - q_t^{e,\mathrm{IK}} \right\|_2^2, \\
\mathcal{L}_{\mathrm{geo}} &= \sum_{t=1}^{T} \sum_{c \in \mathcal{P}_t} [-d_{t,c}]_+^2 \\
&\quad + \beta_{\mathrm{geo}} \sum_{t=1}^{T} \sum_{u \in \mathcal{V}_t} [\delta - s_{t,u}]_+^2, \\
\mathcal{L}_{\mathrm{contact}} &= \sum_{t=1}^{T} \sum_{k \in \mathcal{C}_t} w_{t,k} \left\| p_k^e(q_t^e) - a_{t,k} \right\|_2^2, \\
\mathcal{L}_{\mathrm{temp}} &= \sum_{t=2}^{T} \left\| q_t^e - q_{t-1}^e \right\|_2^2.
\end{aligned}$$

Geometry Loss. The first term of $\mathcal{L}_{\mathrm{geo}}$ detects coarse penetration with the simplified MuJoCo collision geometry: $\mathcal{P}_t$ contains collision pairs with signed distance $d_{t,c}$, where $d_{t,c} < 0$ indicates penetration. The second term checks the robot and object visual meshes to capture fine-scale penetration. Here, $\mathcal{V}_t$ contains samples from the robot visual mesh near the object, and $s_{t,u}$ is the signed distance from sample u to the object surface. This term penalizes both negative $s_{t,u}$, which indicates penetration, and positive distances smaller than the clearance threshold $\delta = 0.005\,\mathrm{m}$. The weight β_{geo} balances this fine-scale visual-mesh penalty with the MuJoCo collision penalty.

Contact Loss. We construct the valid-contact set $\mathcal{C}_t$ from the reconstructed human fingertips. At each frame, fingertip k is transformed into the object coordinate frame, and $\rho_{t,k}$ is its Euclidean distance to the nearest point on the reconstructed object surface. Contact starts when $\rho_{t,k} \le \tau_k$. We select $\tau_k \in [0.02\,\mathrm{m}, 0.06\,\mathrm{m}]$ according to the target embodiment and contact geometry. Once active, a contact remains active until $\rho_{t,k} > \tau_k + 0.01\,\mathrm{m}$ where we use a $1\,\mathrm{cm}$ wider exit threshold to prevent contact labels from toggling when the measured distance fluctuates near τ_k. We discard contact segments shorter than 10 consecutive frames. For each $k \in \mathcal{C}_t$, $a_{t,k}$ is that nearest object-surface point, $w_{t,k} \ge 0$ is the contact-confidence weight obtained from the temporally filtered contact segment, and $p_k^e(q_t^e)$ is the corresponding robot-fingertip position.

B 3D Robot-State Decoder Details

Decoder Architecture. A shared ResNet-34 (He et al. 2016) feature pyramid extracts local and global features. For each robot hand, a palm-anchor head predicts heatmaps and visibility for eight predefined 2D palm anchors, a spatial fingertip heatmap head localizes up to five fingertips, and an MLP state head predicts palm-frame 3D fingertip coordinates and normalized joint configurations. A shared camera head predicts intrinsics. SQPnP (Terzakis and Lourakis 2020) recovers the camera-space wrist pose from the predicted 2D palm anchors and predefined 3D palm geometry. A temporal Transformer refines the framewise states over all 81 frames,

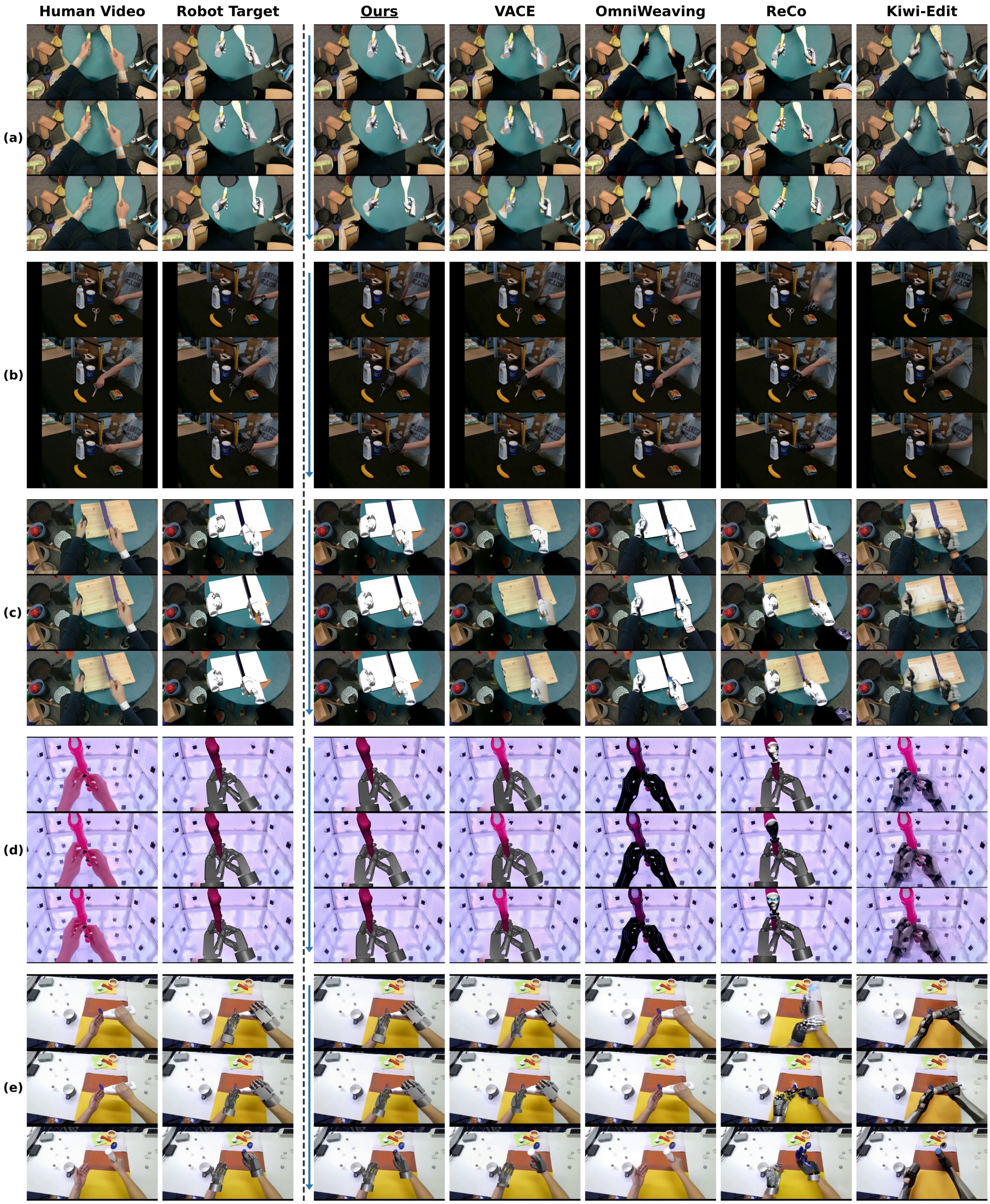

Figure 9: Additional qualitative comparisons on five human-object interactions.

Source dataset	Paired clips	Paired frames	Duration (h)
DexYCB	67,200	5,443,200	50.40
GigaHands	11,715	948,915	8.79
H2O	7,955	644,355	5.97
HOT3D	35,591	2,882,871	26.69
TACO	52,086	4,218,966	39.06
Total	174,547	14,138,307	130.91

Table 4: RoboEdit-14M source distribution, with duration computed at 30 FPS.

and embodiment-specific forward kinematics produces the complete camera-space robot-hand trajectory.

Loss Functions. The decoder is trained with three objectives:

$$\begin{aligned}
\mathcal{L}_{\text{anchor}} &= \lambda^a_{\text{hm}}\mathcal{L}^a_{\text{hm}} + \lambda^a_{\text{uv}}\mathcal{L}^a_{\text{uv}} + \lambda^a_{\text{vis}}\mathcal{L}^a_{\text{vis}},\\
\mathcal{L}_{\text{spa}} &= \lambda_{\text{2D}}\mathcal{L}_{\text{2D}} + \lambda^s_{\text{palm}}\mathcal{L}_{\text{palm}} + \lambda_{\text{cam}}\mathcal{L}_{\text{cam}}\\
&\quad + \lambda^s_{\text{qpos}}\mathcal{L}_{\text{qpos}} + \lambda_K\mathcal{L}_K,\\
\mathcal{L}_{\text{tmp}} &= \lambda^t_{\text{palm}}\mathcal{L}_{\text{palm}} + \lambda^t_{\text{qpos}}\mathcal{L}_{\text{qpos}} + \lambda_{\Delta\text{palm}}\mathcal{L}_{\Delta\text{palm}}\\
&\quad + \lambda_{\Delta\text{qpos}}\mathcal{L}_{\Delta\text{qpos}} + \lambda_{\text{res}}\mathcal{L}_{\text{res}}.
\end{aligned}$$

Here, each λ is a scalar weight for its supervision term; a, s, and t distinguish anchor, spatial, and temporal supervision, respectively. The individual terms are defined as follows:
Anchor Heatmap Loss: $\mathcal{L}^a_{\text{hm}}$ uses a heatmap-distribution loss to supervise the rigid palm-anchor heatmaps.
Anchor Coordinate Loss: $\mathcal{L}^a_{\text{uv}}$ uses Smooth-L1 loss on the 2D image coordinates of the palm anchors.
Anchor Visibility Loss: $\mathcal{L}^a_{\text{vis}}$ uses binary cross-entropy to supervise anchor visibility.
2D Fingertip Loss: $\mathcal{L}_{\text{2D}}$ combines a heatmap-distribution loss with Smooth-L1 loss on the 2D fingertip coordinates.
3D Geometry Losses: $\mathcal{L}_{\text{palm}}$ and $\mathcal{L}_{\text{cam}}$ use Smooth-L1 losses on 3D fingertip coordinates in the palm and camera frames, respectively.
Joint-State Loss: $\mathcal{L}_{\text{qpos}}$ uses Smooth-L1 loss on normalized robot joint states.
Camera-Intrinsics Loss: $\mathcal{L}_K$ uses Smooth-L1 loss on the predicted camera intrinsics.
Temporal Motion Losses: $\mathcal{L}_{\Delta\text{palm}}$ and $\mathcal{L}_{\Delta\text{qpos}}$ use Smooth-L1 losses to match adjacent-frame fingertip and joint-state motion.
Temporal Residual Loss: $\mathcal{L}_{\text{res}}$ applies an L2 penalty to the temporal corrections.

Unavailable fingertips, joints, and frames are excluded using validity masks. $\mathcal{L}_{\text{anchor}}$, $\mathcal{L}_{\text{spa}}$, and $\mathcal{L}_{\text{tmp}}$ are the weighted combinations shown above.

C RoboEdit-14M Statistics

Tables 4 and 5 summarize RoboEdit-14M by source and target embodiment, while Fig. 10 shows representative synthetic human/robot pairs.

D Training Details

Computing Infrastructure. Experiments were conducted on Ubuntu 22.04.5 workstations equipped with two AMD EPYC 9354 CPUs, 1.5 TiB of host memory, and eight NVIDIA H100 NVL GPUs, each with 94 GiB of memory.

Embodiment	Real	Synthetic	Total
Ability	24,197	4,839	29,036
Allegro	24,197	4,839	29,036
Unitree Dex3	12,237	2,447	14,684
Inspire	24,197	4,839	29,036
Panda gripper	12,237	2,447	14,684
SCHUNK SVH	24,197	4,838	29,035
XHand	24,197	4,839	29,036
Total	145,459	29,088	174,547

Table 5: RoboEdit-14M distribution by target embodiment.

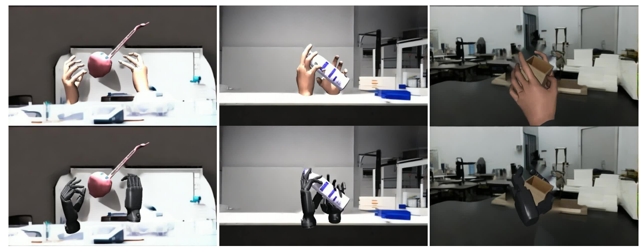

Figure 10: Synthetic paired-video examples. Each column shows an aligned synthetic human frame and its robot target under the same scene, object state, and camera view.

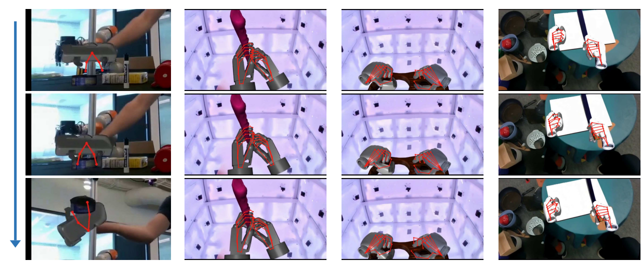

Figure 11: Additional 3D Robot-State Decoder results on RoboEdit-Trans edited videos. Red overlays denote predicted camera-space hand states.

Backbone and Cross-Embodiment Adaptation. We first train the video-editing backbone for approximately 10K steps, updating 283.4M parameters with AdamW at a learning rate of 5×10^{-5} and an effective batch size of 12 across eight GPUs. We then freeze the backbone and jointly train LoRA adapters on selected attention and feed-forward projections together with residual bottleneck adapters in the final ten transformer blocks, using an effective batch size of 16 across four GPUs. The LoRA and residual-adapter branches contain 2.5M and 7.9M trainable parameters, respectively.

3D Robot-State Decoder. We first train the palm detector and framewise spatial estimator on rendered and composited robot frames. We then freeze both components and train temporal refinement on complete 81-frame sequences. We train the spatial and temporal stages for approximately 8.9K and 1.5K optimization steps, respectively; both stages use

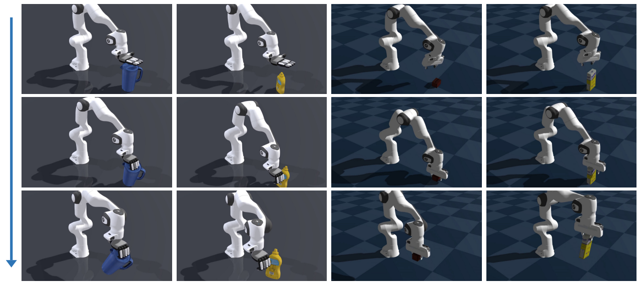

Figure 12: Simulation rollouts of the trajectory-conditioned controller across four YCB-object manipulation tasks.

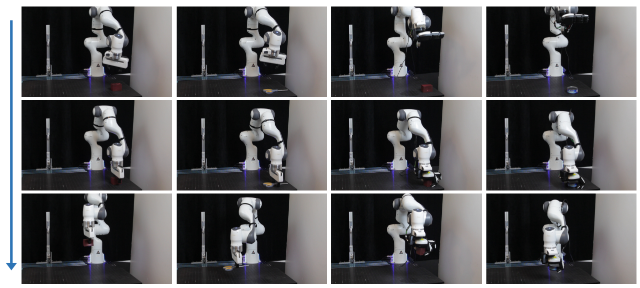

Figure 13: Additional real-robot deployment results across four YCB-object tasks.

AdamW with a learning rate of 2×10^{-5}.

Qwen-Image-Edit. We fine-tune Qwen-Image-Edit for approximately 4.9K optimization steps on aligned image pairs sampled from RoboEdit-14M. At inference, it generates robot references at indices $\{0, 10, \ldots, 80\}$, after which RoboEdit-Trans produces an 81-frame video.

Randomness Control. For reproducibility, we fix the random seeds used for data sampling, decoder training, and model inference. RoboEdit-Trans data sampling initializes its NumPy generator with 20260531 plus the epoch index. Decoder training initializes the Python, NumPy, and PyTorch generators with 20260705 plus the distributed rank. RoboEdit-Trans, Qwen-Image-Edit, and baseline inference use deterministic per-sample PyTorch seeds derived from 42, 20260629, and 20260624, respectively.

E Trajectory-Conditioned Control and Real-Robot Deployment

We evaluate whether the robot-hand trajectories recovered by the 3D Robot-State Decoder are physically executable and can supervise downstream control. We first train a trajectory-conditioned controller in simulation and then execute simulation-validated trajectories on a physical Franka Panda.

Trajectory-conditioned residual control. For each interaction, the 3D Robot-State Decoder provides a decoded robot-hand trajectory $\tau^h = \{h_t^\star\}_{t=1}^T$, which serves as the control reference; $h_t^\star$ denotes the reference palm or end-effector state. We represent the trajectory relative to its first frame and resample it to the control horizon. Given the current task state s_t and the robot-hand tracking error $h_t^\star - p_t$, the residual policy π_ϕ predicts a bounded correction to a nominal joint command computed by inverse kinematics (IK):

$$a_t = \pi_\phi(s_t, h_t^\star - p_t)\,,$$
$$q_t^{\text{cmd}} = \text{clip}\big(q_t^{\text{IK}} + \rho_{\text{act}} a_t, q_{\min}, q_{\max}\big)\,.$$

Here, π_ϕ is the residual policy with parameters ϕ, s_t is the current robot and object task state, and p_t is the current palm or end-effector state. q_t^{IK} is the nominal arm configuration computed by IK, and q_t^{cmd} is the resulting command sent to the low-level arm controller; $a_t \in [-1, 1]^7$ is the residual arm action, and $\rho_{\text{act}} = 0.02\,\text{rad}$ is its scale. $q_{\min}$ and $q_{\max}$ denote the Franka arm joint limits. The operator clip enforces these limits elementwise. The low-level controller executes q_t^{cmd}, whose outcome is evaluated through palm tracking and task success.

Control objective. The reward supervises tracking of the decoded robot-hand trajectory while encouraging successful and smooth task execution:

$$r_t = w_{\text{hand}} \exp(-\beta \, \|p_t - h_t^\star\|_2) + r_t^{\text{task}} - w_{\text{act}} \, \|a_t\|_2^2\,.$$

Here, r_t is the total reward; w_{hand} and w_{act} weight trajectory tracking and action regularization, respectively; β controls sensitivity to the Euclidean tracking error; and r_t^{task} contains rewards for grasp maintenance, object-motion consistency, and lift completion. The decoder trajectory therefore conditions both the policy action and its training objective.

Simulation setup. We train the residual policy using PPO (Schulman et al. 2017) in Genesis (Genesis Authors 2024), with 512 parallel environments spanning 18 YCB objects. A single policy is trained across all objects, without per-object fitting. During training, the initial object position and yaw are randomized by up to 4 cm and 23°, respectively.

A rollout succeeds if the object remains grasped and its terminal position is within 8 cm of the demonstrated target. Across the randomized environments, the controller achieves trajectory-reproduction success rates of 71% for the Panda gripper and 62% for XHand. Representative simulation rollouts are shown in Fig. 12.

Real-robot deployment. We deploy the controller on a 7-DoF Franka Panda, using the decoded robot-hand trajectories as control references for YCB-object manipulation.

F Additional Qualitative Results

Figure 9 extends the main-paper comparison with five additional interaction examples. We again use midpoint non-keyframes at indices 5, 35, and 65. Figure 11 shows additional decoded robot states from edited videos, and Figure 13 presents additional real-robot results on YCB-object manipulation with both end effectors.

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